Mastering Generative AI & LLM Engineering
From Foundations to Applications

Large Language Models

Lecturer: Chetan Gupta
in association with
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Topics to be covered

- Introduction
- Transformers
- GPT
- HuggingFace
- BERT
- XLNeT

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Common LLMs

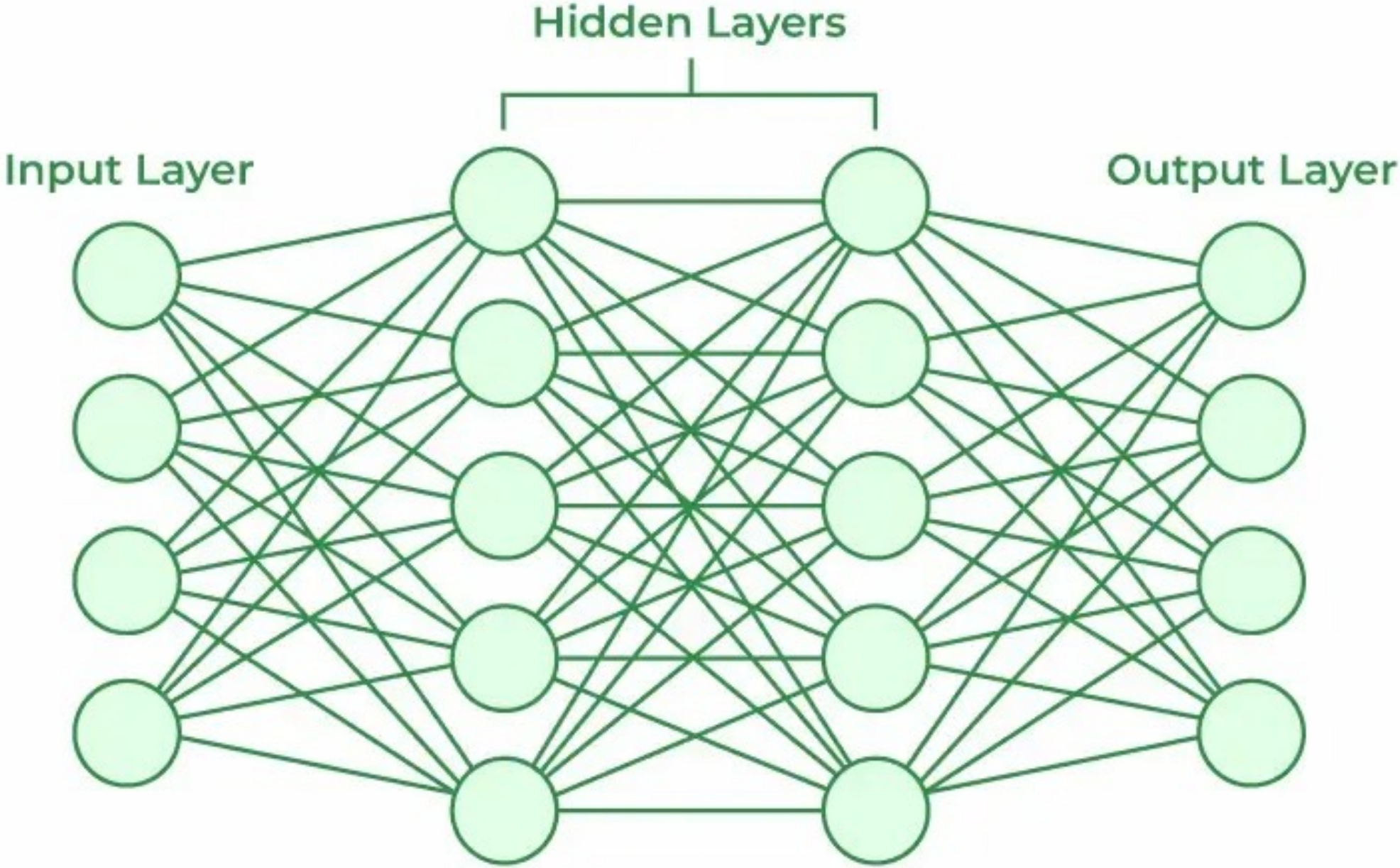
- Bard
- Dolly
- Llama
- BLOOM
- Alpaca
- Cohere
- Wizard

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- Write Poetry
- Write Essays
- Create Social Media Content
- Write Code
- ..and so on




Hidden Layers
Input Layer
Output Layer

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Transformers

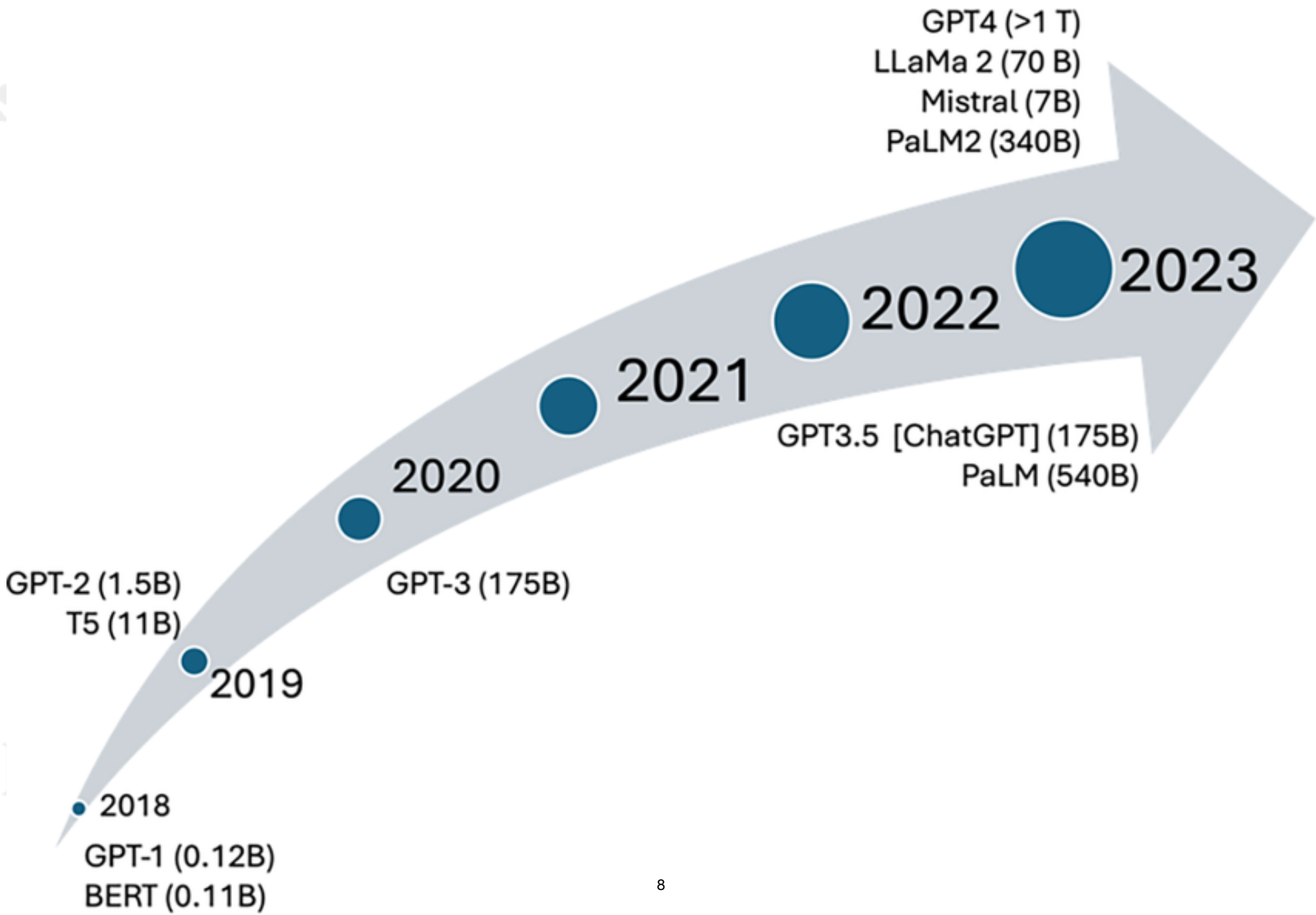
Main Features to LLMs :

- LLMs are ‘large’
- General Purpose
- Be pre-trained and fine-tuned



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How Large ?


GPT4 (>1 T)
LLaMa 2 (70 B)
Mistral (7B)
PaLM2 (340B)
2023
2022
2021
GPT3.5 [ChatGPT] (175B)
PaLM (540B)
2020
GPT-3 (175B)
GPT-2 (1.5B)
T5 (11B)
2019
2018
GPT-1 (0.12B)
BERT (0.11B)

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More Text = More successful at learning

Text data include : Books, Websites, Wikipedia, Articles, Poems, etc.

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“General Purpose”

1. Trained on a variety of text data
2. Designed to understand and generate human language

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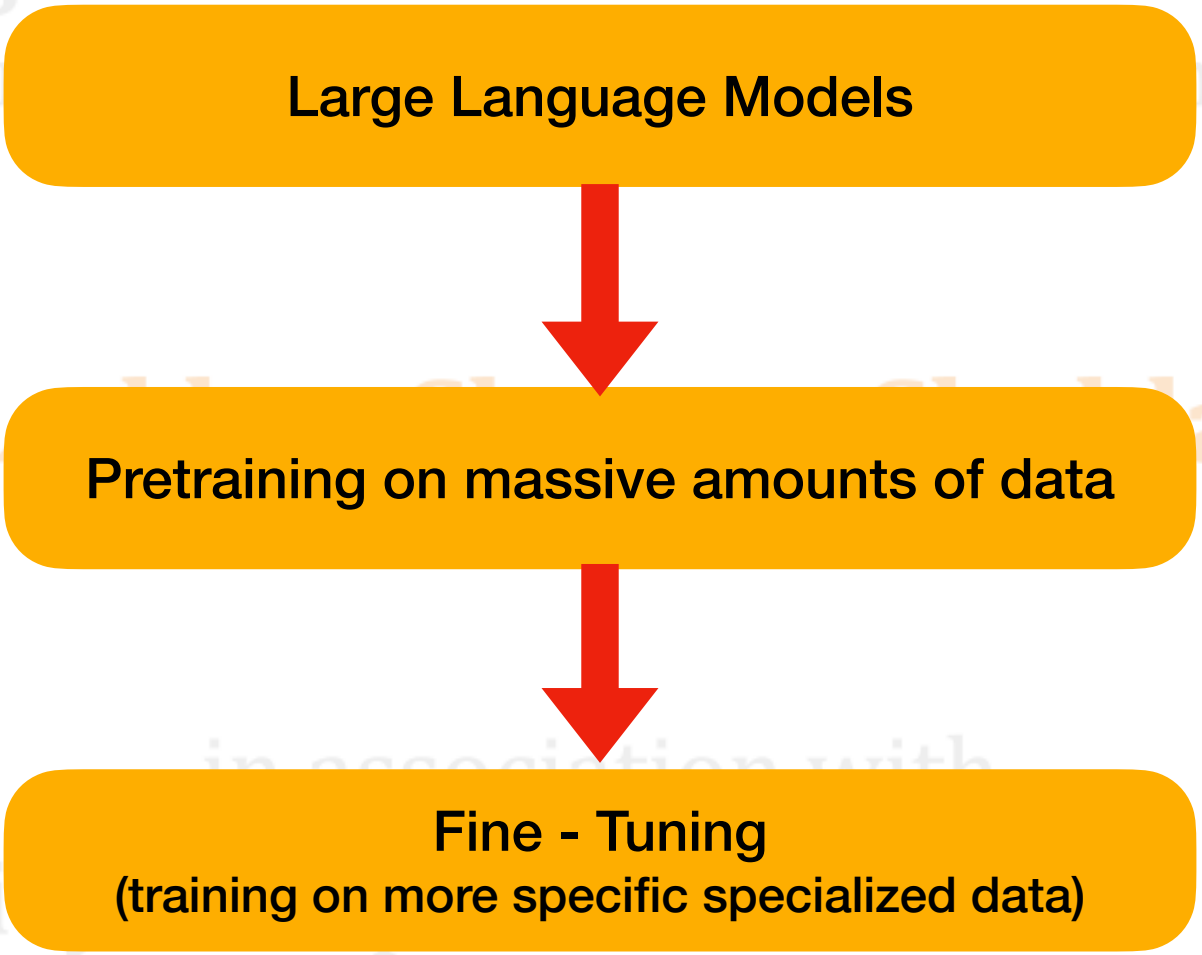
LLMs’ goal : Give general understanding of knowledge and how language works so it can be applied to more specific tasks later on.

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Fine - Tuning

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Few Shot : Minimal Data to train a model

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Zero Shot : Model can recognize things that it has not been explicitly taught in the training

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iris setosa



iris versicolor



iris virginica



I have a flower with many petals and thorns on the stem. Which flower could this be?

Most likely a **rose** — that's the classic flower with lots of petals and (prickly) thorns on the stem.

How I tell:

- Thorns/prickles on the stem are a strong rose clue.
- Many-petaled blooms (especially double/old garden or hybrid-rose varieties) match your description.
- Roses often have a strong scent and pinnate leaves with serrated edges.

Other possibilities (less likely): **peonies** and **ranunculus** also have many petals but *don't* have thorns.

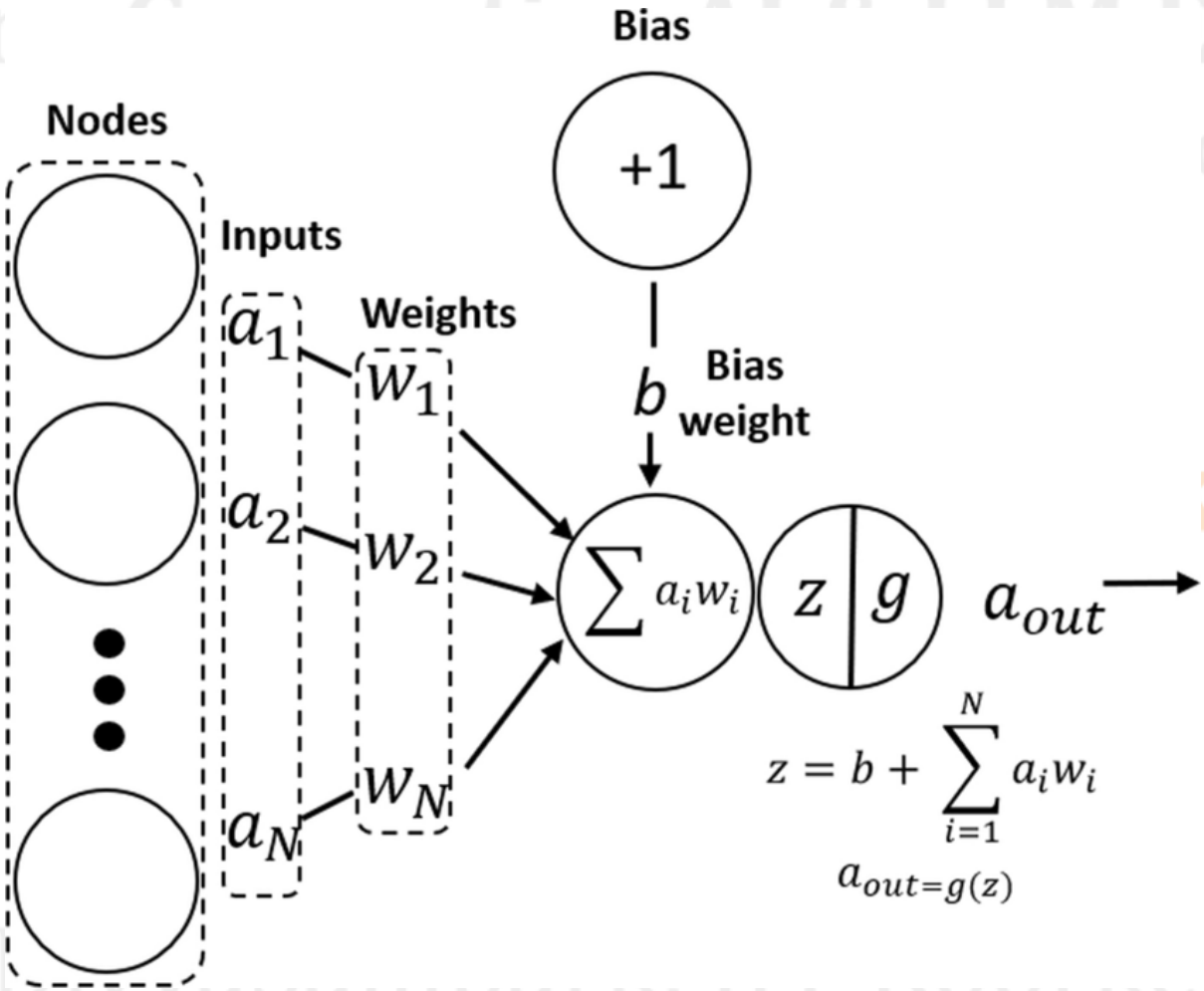
If you want a definite ID, you can upload a photo and I'll confirm. 🌹

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LLMs are based on Transformer Architecture

Transformers are a type of Neural Network subset of Deep Learning

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Bias
Nodes
+1
Inputs
Weights
a_1
w_1
b
Bias weight
a_2
w_2
$\sum a_i w_i$
z
g
a_{out}
w_N
a_N
$z = b + \sum_{i=1}^{N} a_i w_i$
$a_{out=g(z)}$

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Different Neural Network for different problems

Convolutional Neural Networks

- Vision Classification
- Analyzing Images
- Analyzing Languages

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Different Neural Network for different problems

Recurrent Neural Networks

English	I	am	going	to	buy	a	car
French	Je		vais		acheter	une	voiture

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The Hindu is an Indian English-language daily newspaper. It was first issued in 1878

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RRNs	Transformers
The Hindu is an Indian English-language daily newspaper. It was first issued in 1878	The Hindu is an Indian English-language daily newspaper. It was first issued in 1878

Transformers ‘pay attention’ to the most important words in a piece of text

“Attention is all you need”

Transformers + Attention Mechanism

“enables a model to with the importance of different words or tokens in an input sequence when producing an output”

Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com

Noam Shazeer* Google Brain noam@google.com

Niki Parmar* Google Research nikip@google.com

Jakob Uszkoreit* Google Research usz@google.com

Llion Jones* Google Research llion@google.com

Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu

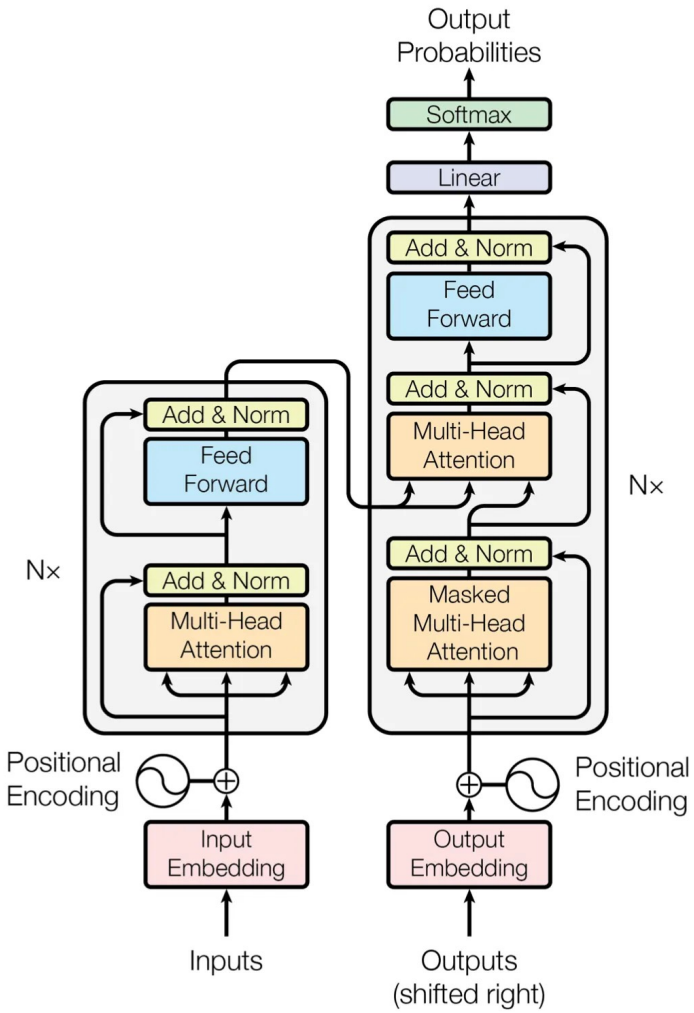
Łukasz Kaiser* Google Brain lukaszkaiser@google.com

Illia Polosukhin* ‡ illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Transformer Architecture



How a Transformer Translates a Sentence

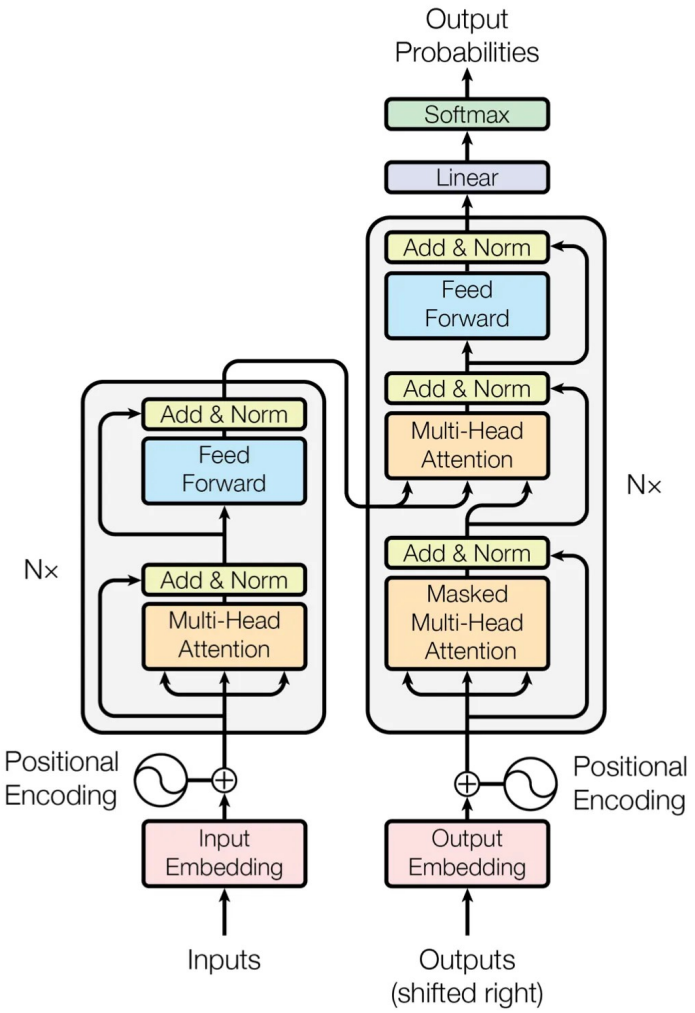
(English → French)

Let's translate this sentence:

English:
"I love machine learning."

Target French translation:
"J'adore l'apprentissage automatique."

Now let's see how a Transformer processes this step-by-step.



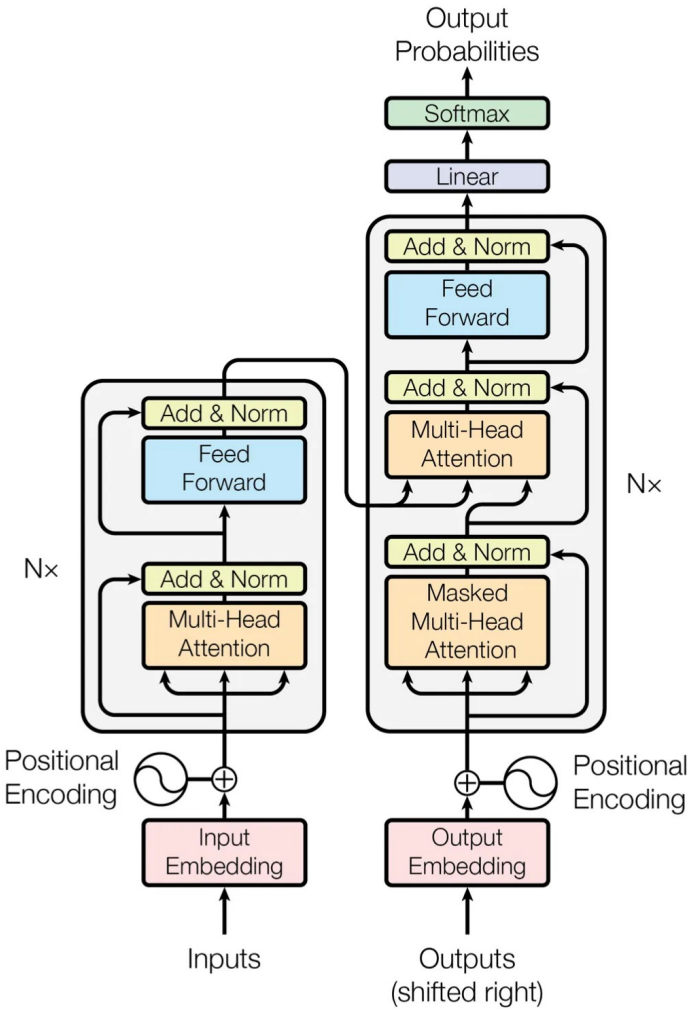
How a Transformer Translates a Sentence

1. Tokenization

The input is broken into tokens:

Input tokens:
["I", "love", "machine", "learning", "."]

The output French tokens are also predefined during training:
["J'", "adore", "l'", "apprentissage", "automatique", "."]



How a Transformer Translates a Sentence

2. Encoder : Understand the English

Each English token is converted into vectors (embeddings + positional encoding).

The encoder processes them through multi-head attention:

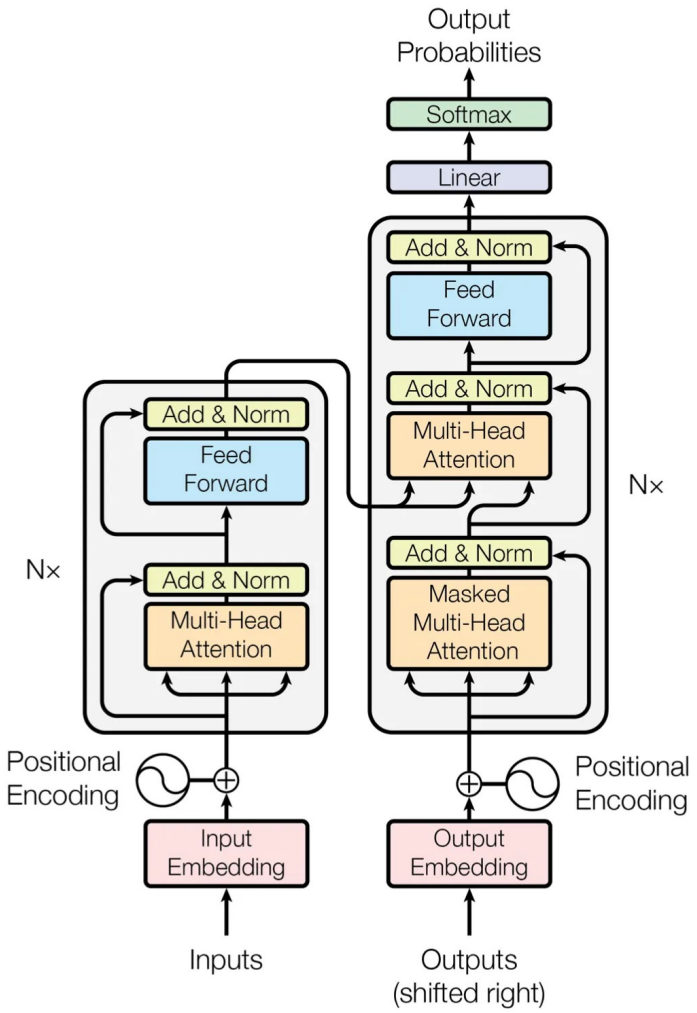
Multi-head attention finds relationships:

“love” attends strongly to “I” → subject-verb relation

“machine” attends to “learning” → phrase grouping

“machine learning” as a whole gets encoded as a concept

The encoder output is a set of enriched vectors that represent the entire meaning of the English sentence.



How a Transformer Translates a Sentence

3. Decoder: Generating French Translation

Now the decoder starts producing French tokens auto-regressively.

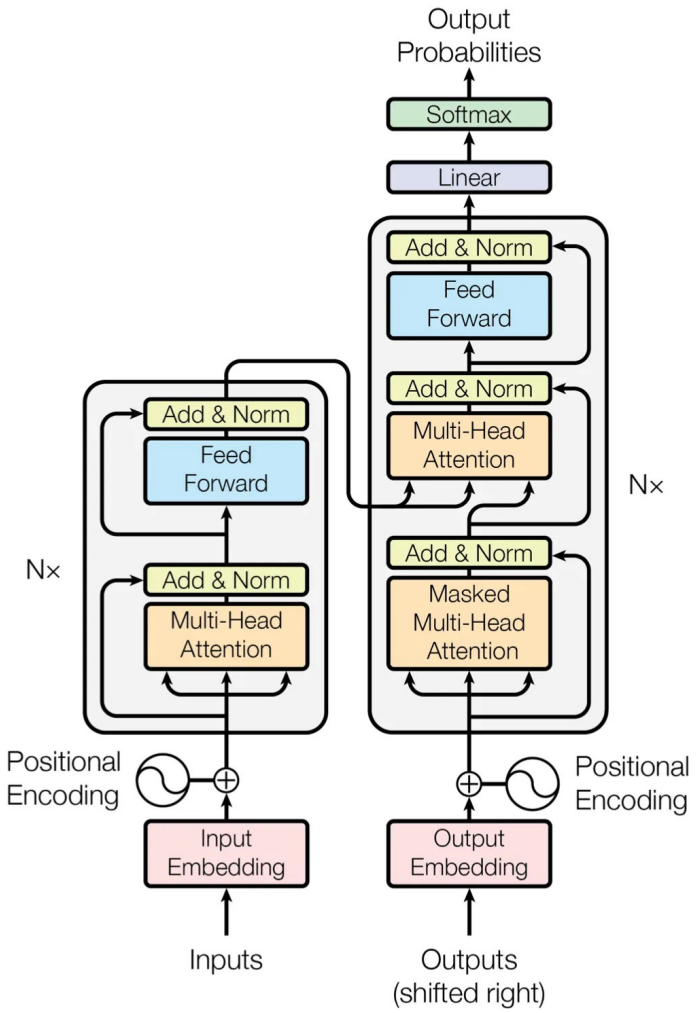
Step 1: Start Token

Decoder receives <start> and asks:

“What should the first word be?”

Using attention over encoder outputs, it chooses:

➡️ “J’”



How a Transformer Translates a Sentence

3. Decoder: Generating French Translation

Step 2: Predict next token

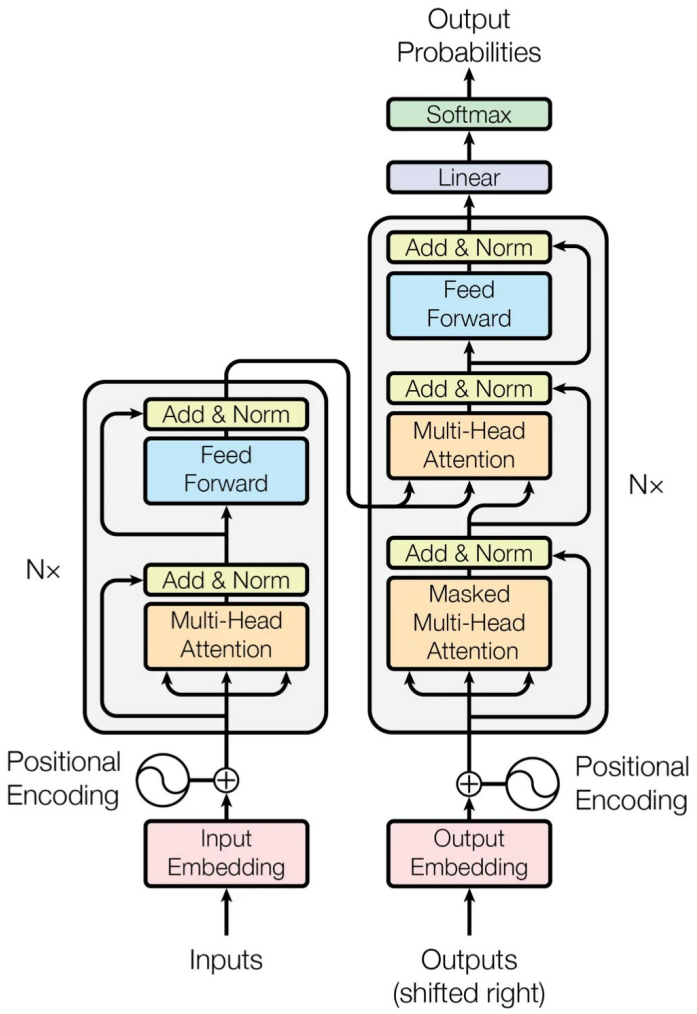
Input to decoder now is: ["J'"]

It looks at: previous tokens (“J’”)

encoder meaning (“I love machine learning”)

Decoder cross-attends to the English embeddings and outputs:

“adore”
because it connects “love” → “adore”.



How a Transformer Translates a Sentence

3. Decoder: Generating French Translation

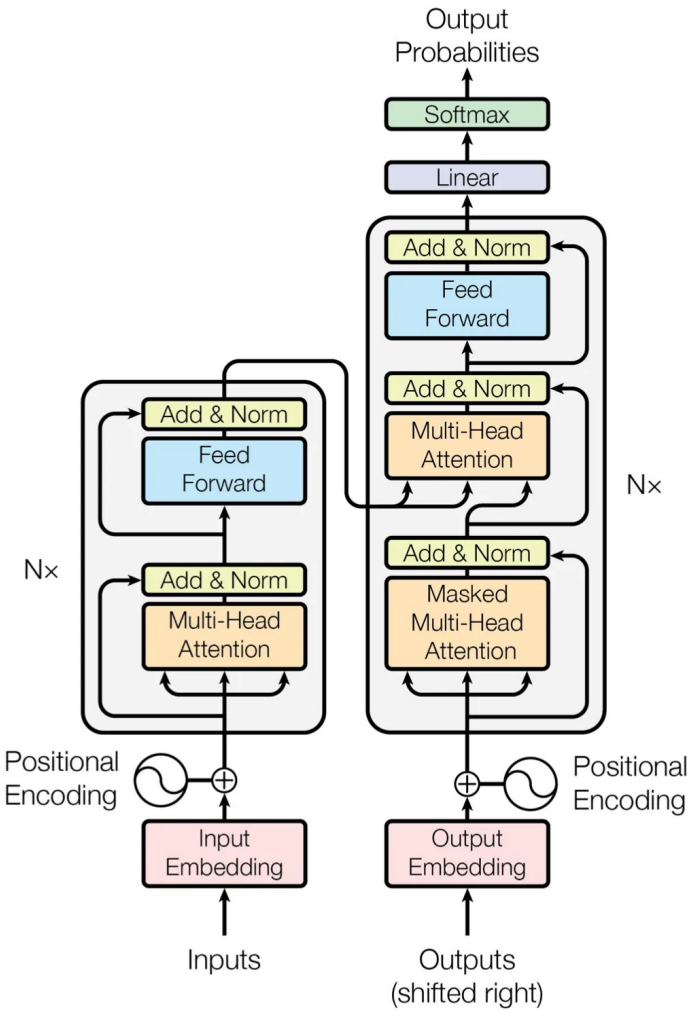
Step 3: Next token

Input now: ["J'", "adore"]

Decoder attends to the English phrase “machine learning.”

It predicts: “l’”

(corresponding to “the” before a vowel in French)



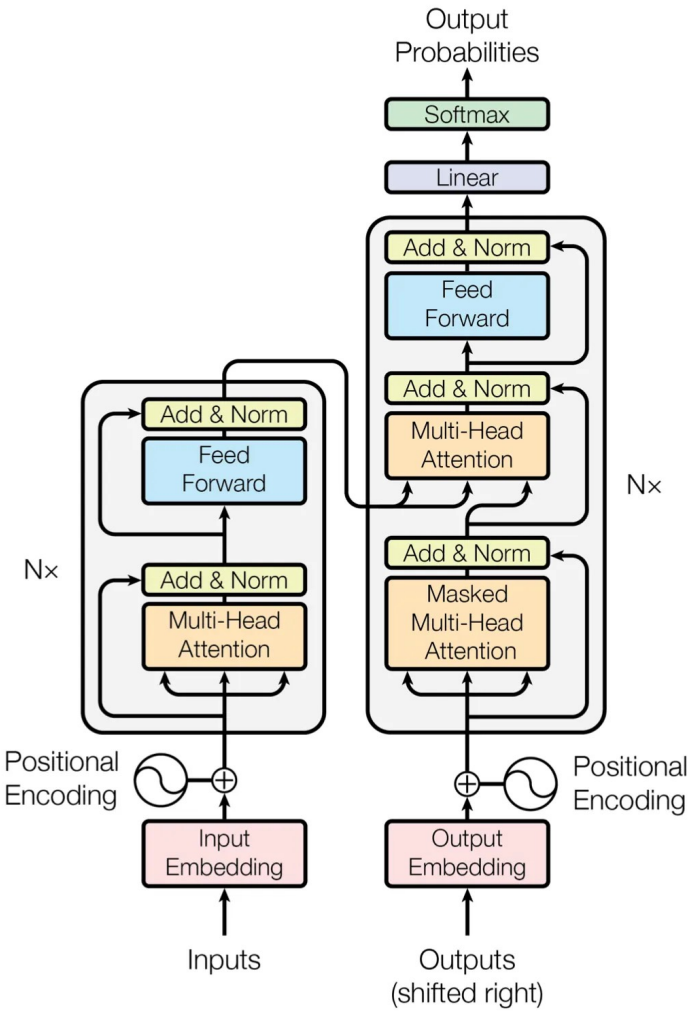
How a Transformer Translates a Sentence

3. Decoder: Generating French Translation

Step 4: Next

Input:
["J'", "adore", "l'"]

Decoder predicts: “apprentissage”



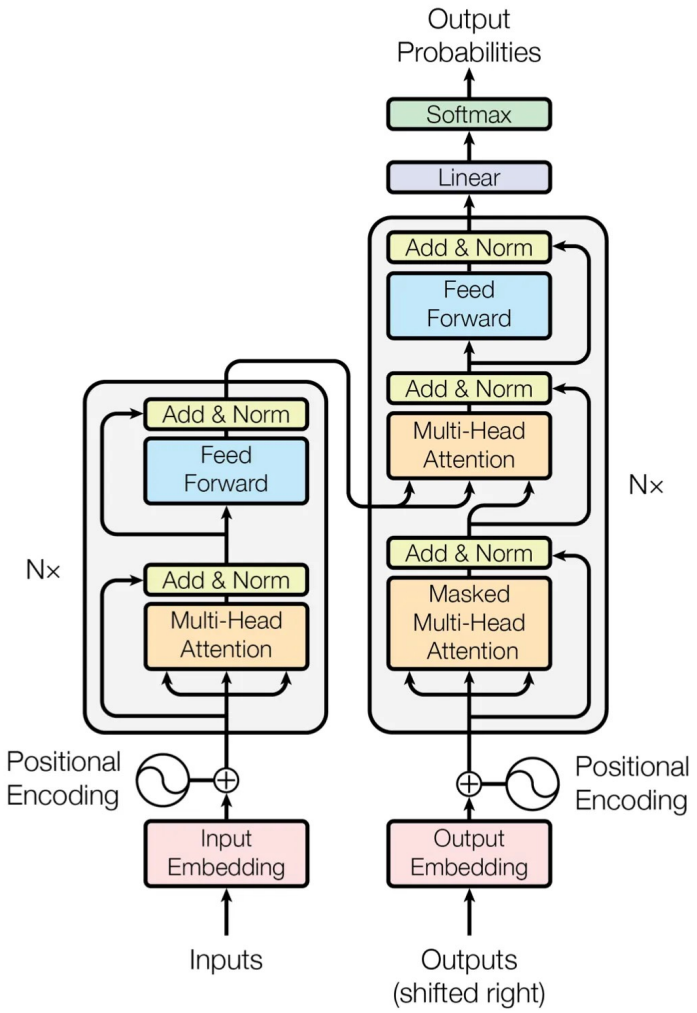
How a Transformer Translates a Sentence

3. Decoder: Generating French Translation

Step 5: End

Input:
["J'", "adore", "l'", "apprentissage"]

Decoder predicts: “automatique”

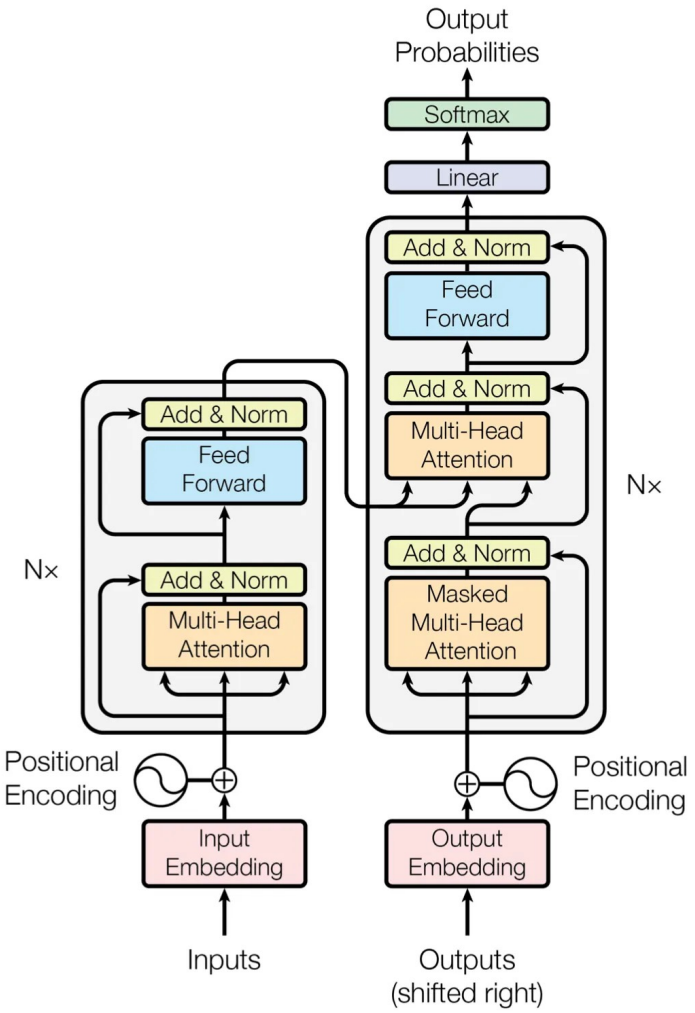


How a Transformer Translates a Sentence

3. Decoder: Generating French Translation

Final output

French: “J’adore l’apprentissage automatique.”



How a Transformer Translates a Sentence

In Simple Terms

1. Encoder

- Understands the whole English sentence at once
- builds contextual meaning.

2. Decoder

- Generates French words one-by-one while

- looking at what it has already generated

- cross-attending the English meaning

- learning grammar + semantics of French.

3. Multi-Head Attention

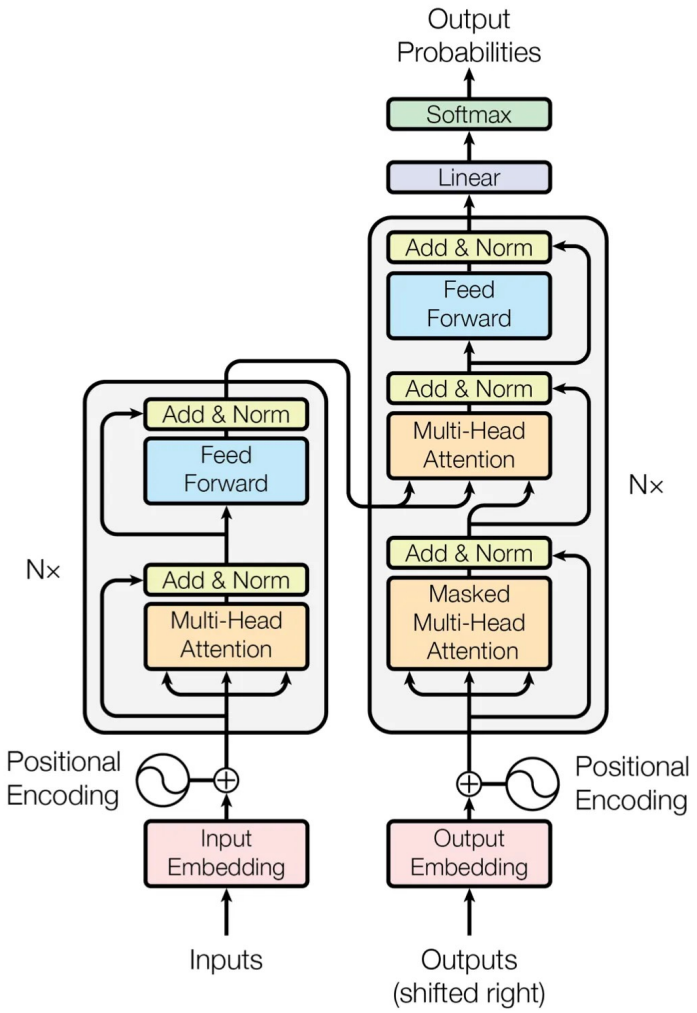
- Allows focusing on different relationships simultaneously:

- subject–verb

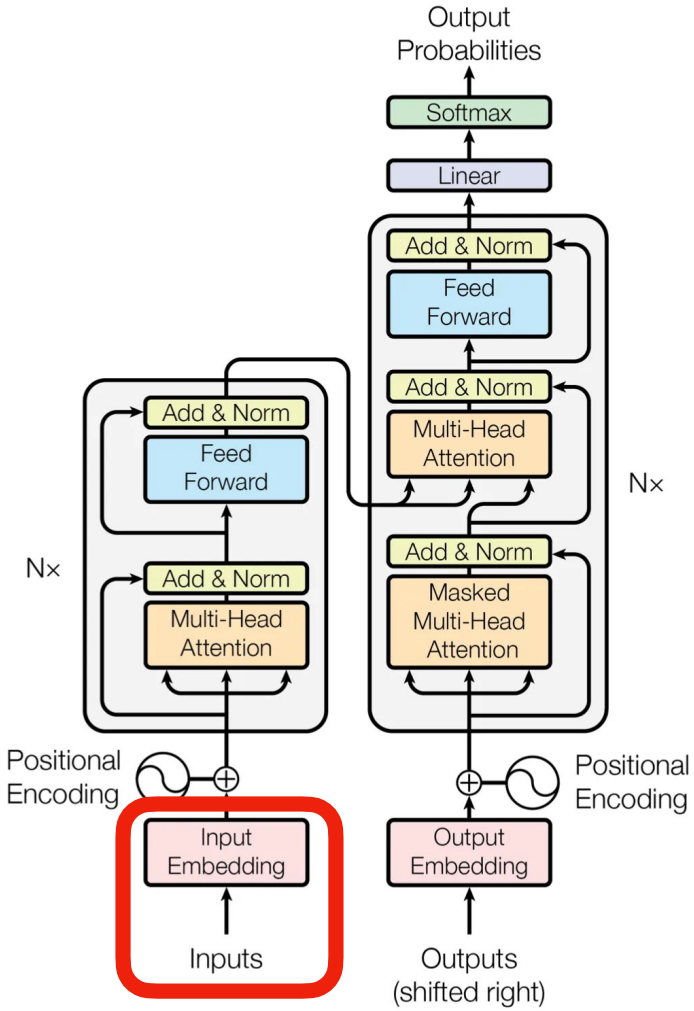
- phrase grouping

- semantic similarity

- long-range dependencies

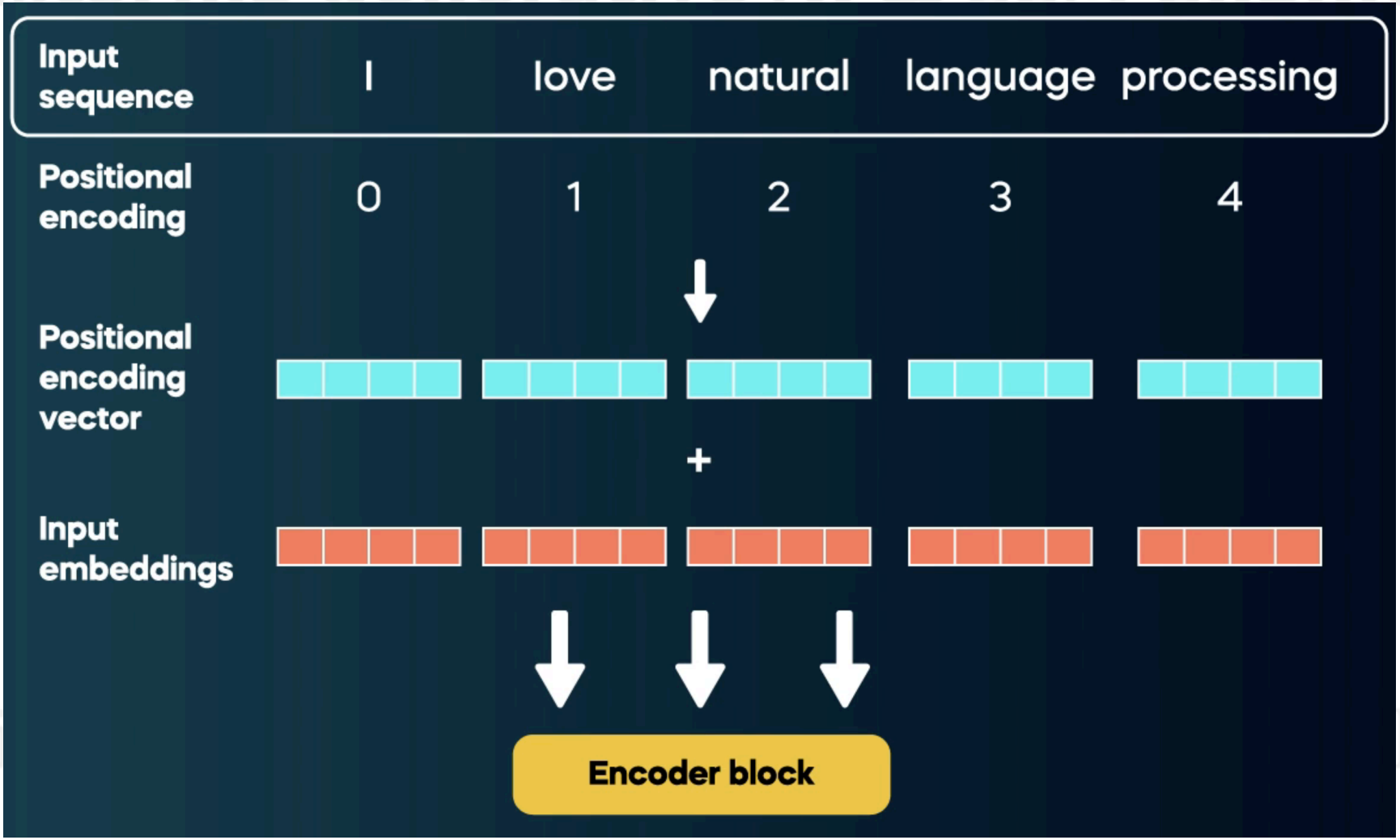


Transformer Architecture

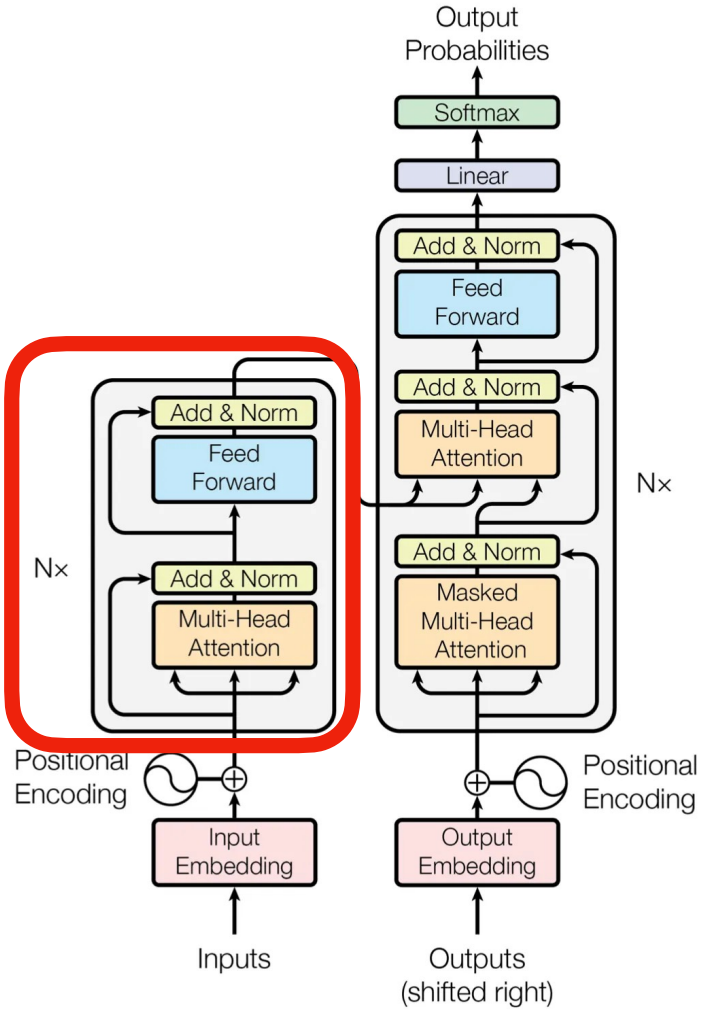


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Positional Encoding



Encoder Block



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How Attention Vectors Are Calculated in Transformers

1. Each token generates three vectors:

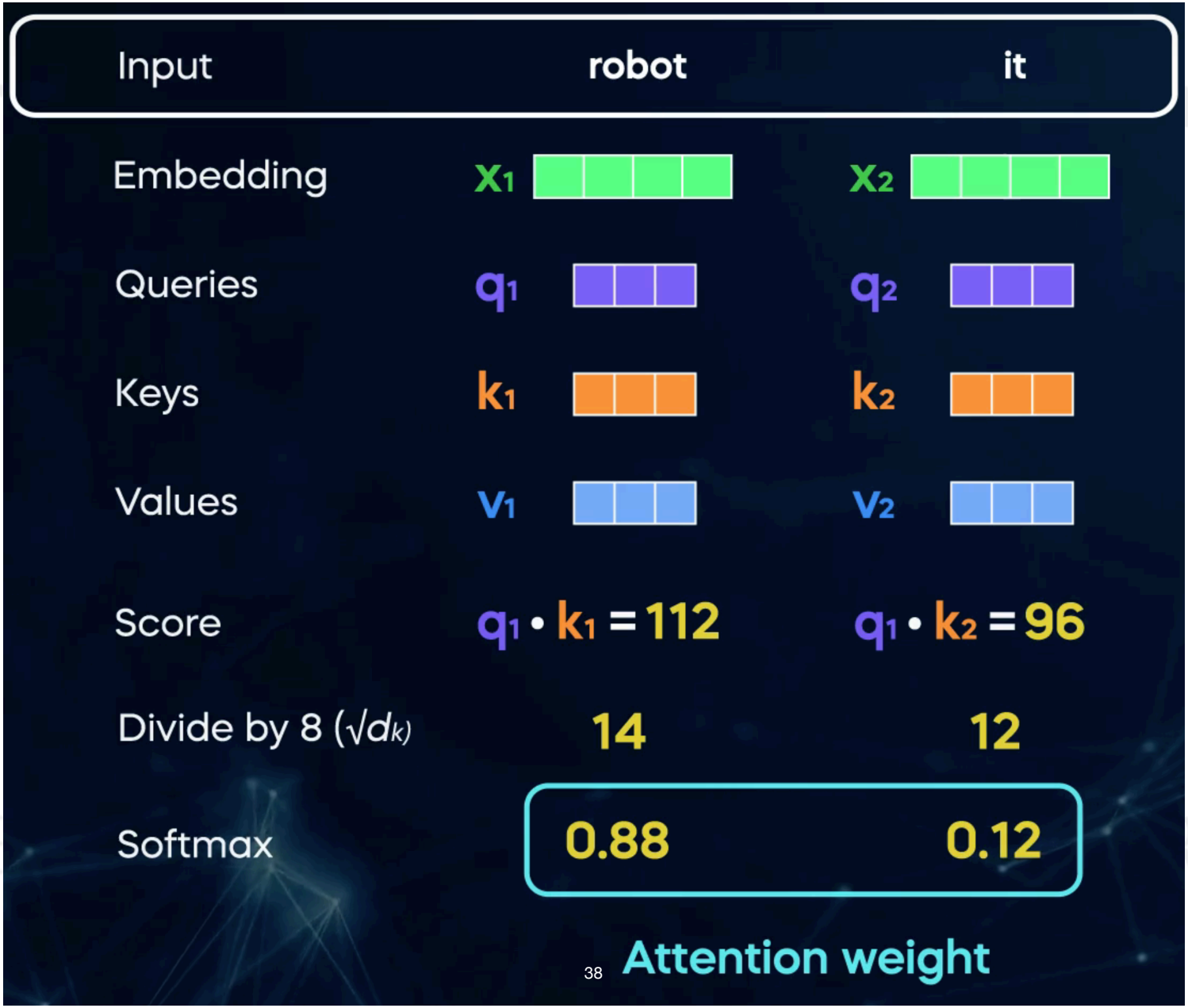
- **Query (Q)** – asks: *"How much should I pay attention to others?"*
- **Key (K)** – tells others *"Here's why you should pay attention to me."*
- **Value (V)** – carries the actual information to be shared.

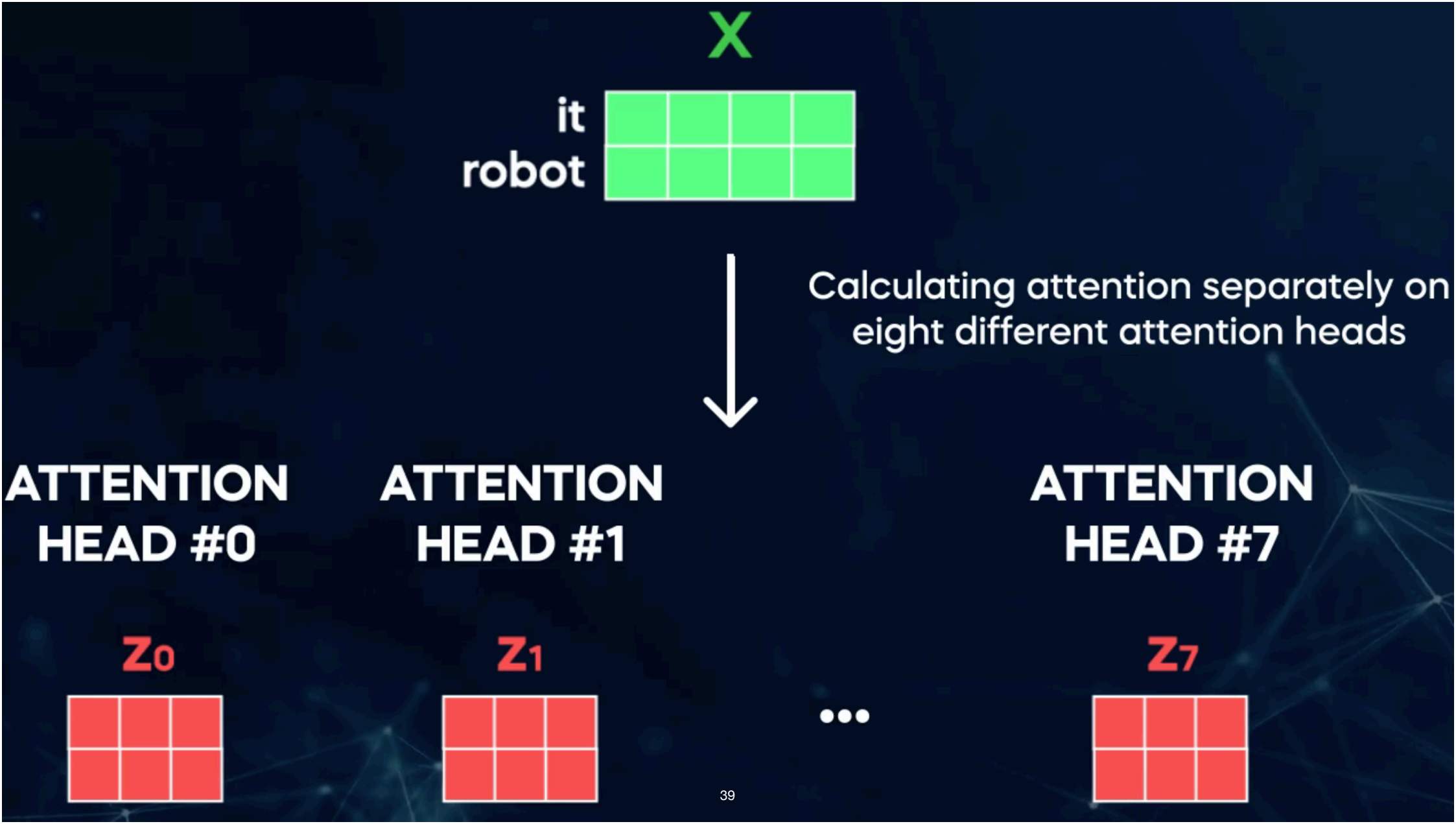
Led by : Shreyas Shukla

2. How they work together:

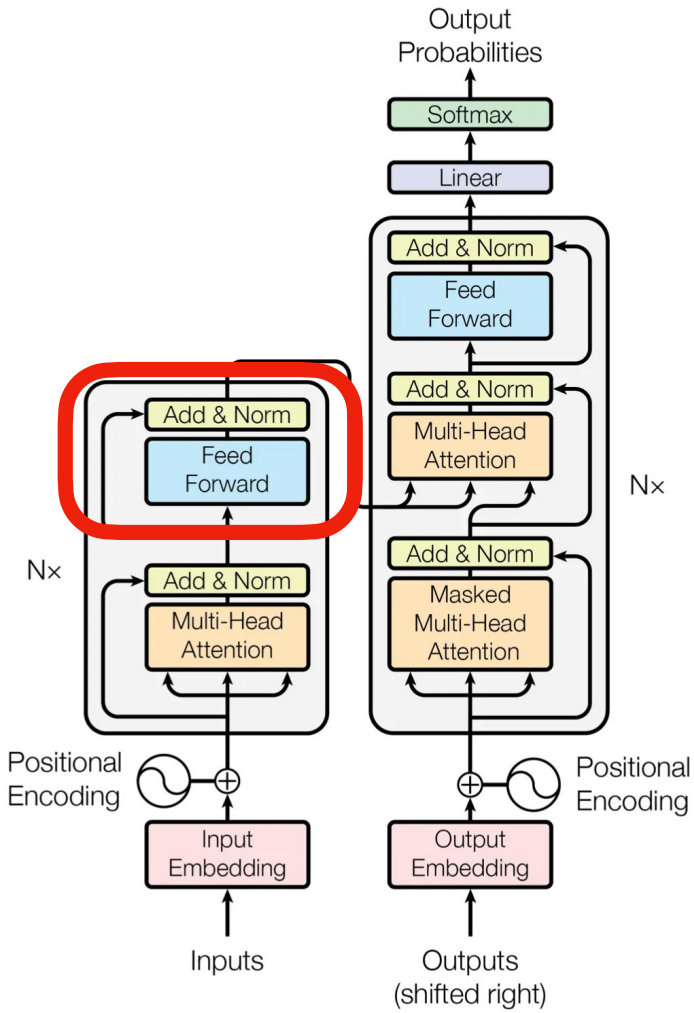
- The **Query** of the current token compares with the **Keys** of all tokens.
- This comparison produces **attention scores** (how relevant each token is).
- The **Values** are then weighted using these scores to form the final attention output.
- Their patterns and meanings are learned during training to optimize performance.

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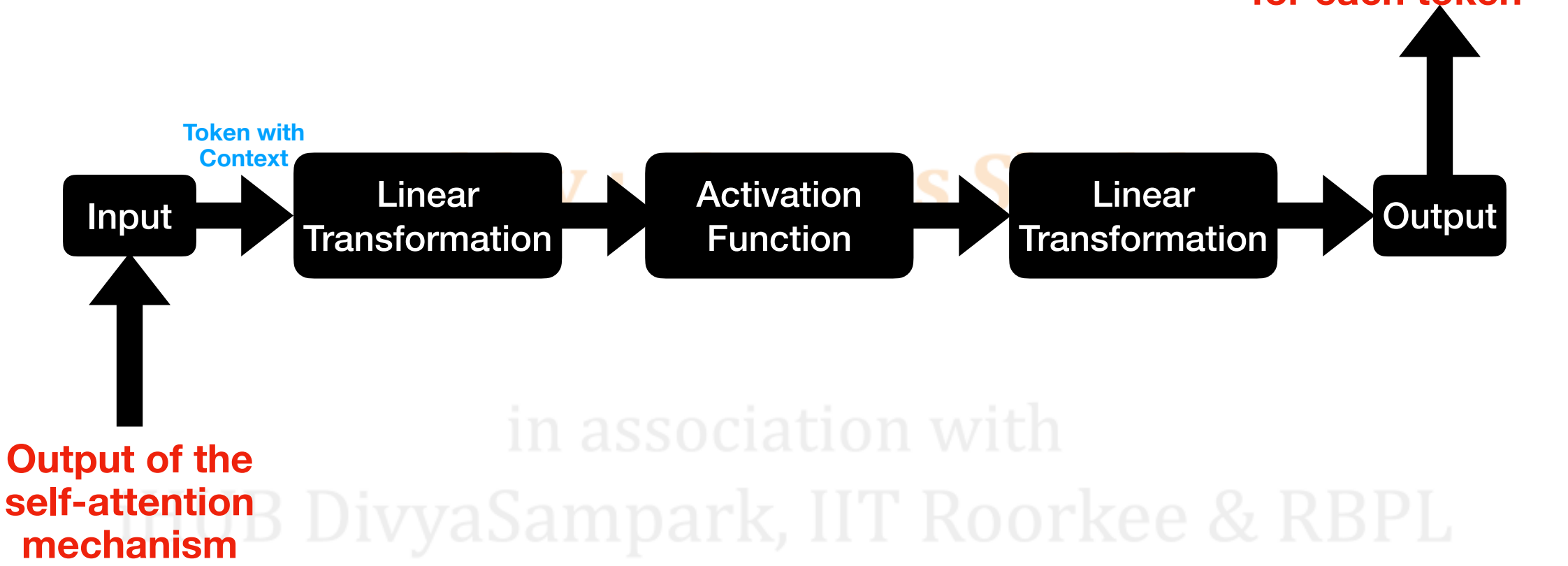

Input
robot
it
Embedding
x1
x2
Queries
q1
q2
Keys
k1
k2
Values
v1
v2
Score
q1 • k1 = 112
q1 • k2 = 96
Divide by 8 (√dk)
14
12
Softmax
0.88
0.12
Attention weight
38


X
it
robot
Calculating attention separately on eight different attention heads
ATTENTION HEAD #0
ATTENTION HEAD #1
ATTENTION HEAD #7
Z0
Z1
Z7

Encoder Block



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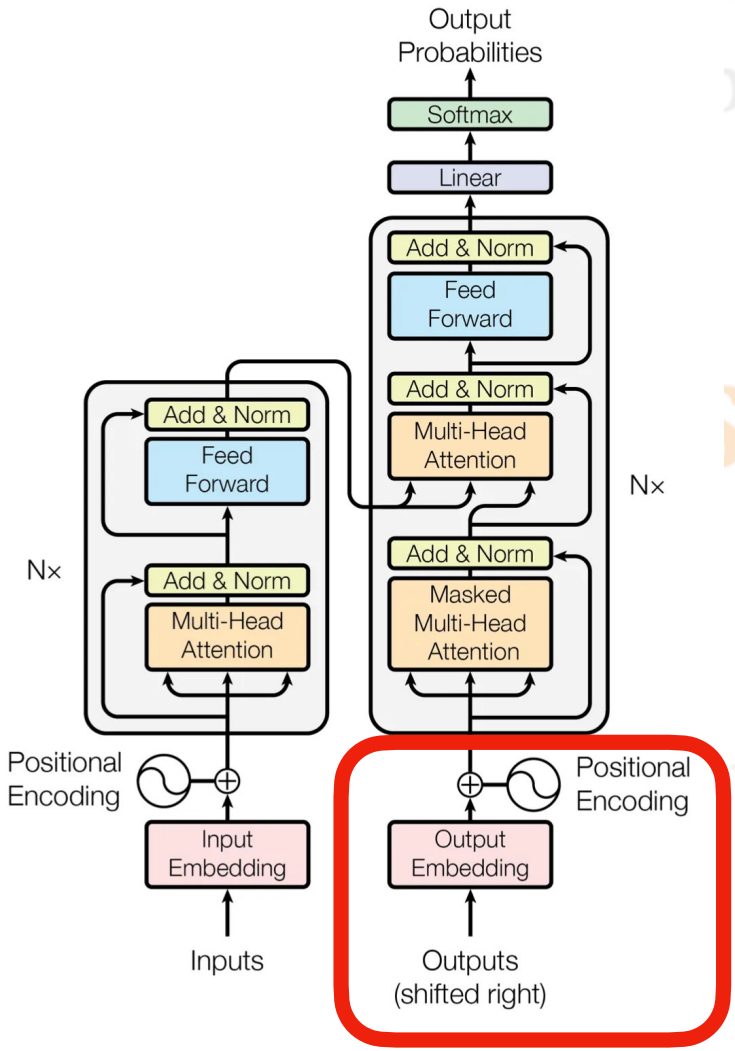
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Feedforward Layer :

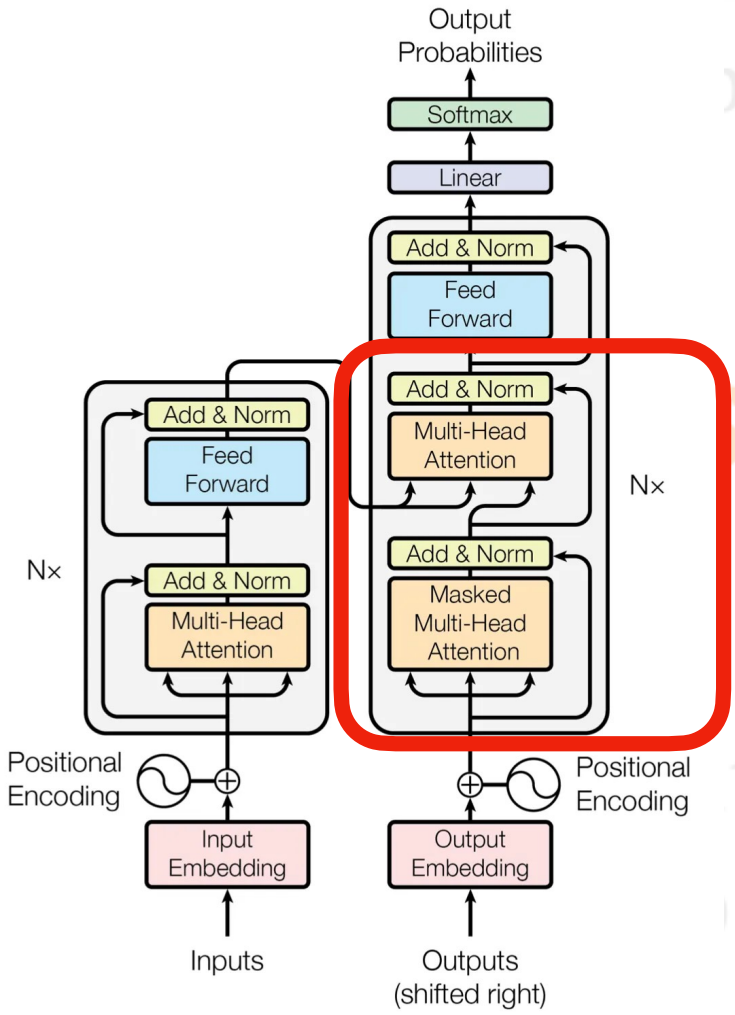
- Applied Independently and can by run in parallel
- A şet of neural network operations applied independently to each token’s representation
- Transformer model capture and encode complex, non-linear relationships between tokens in the input sequence

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Decoder Block



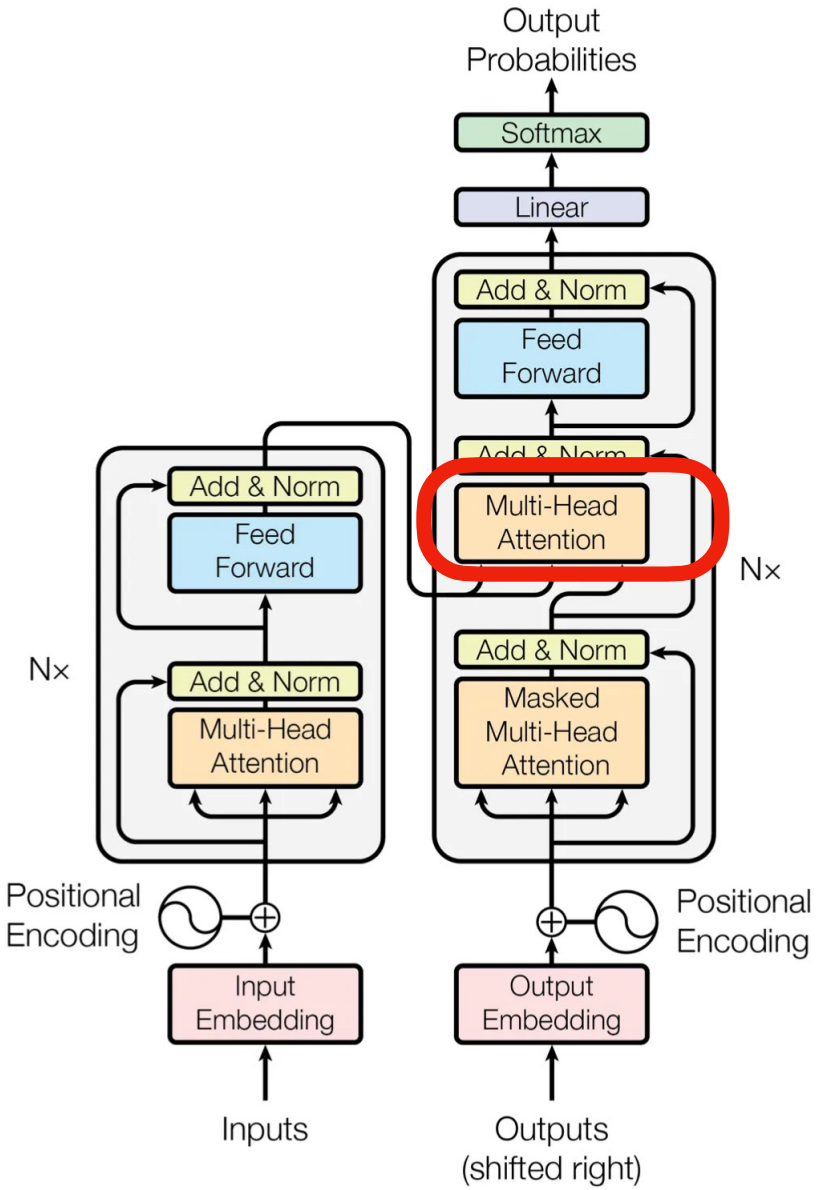
Decoder Block



Multi-Head Attention

1. Calculates attention scores between output token and encoder

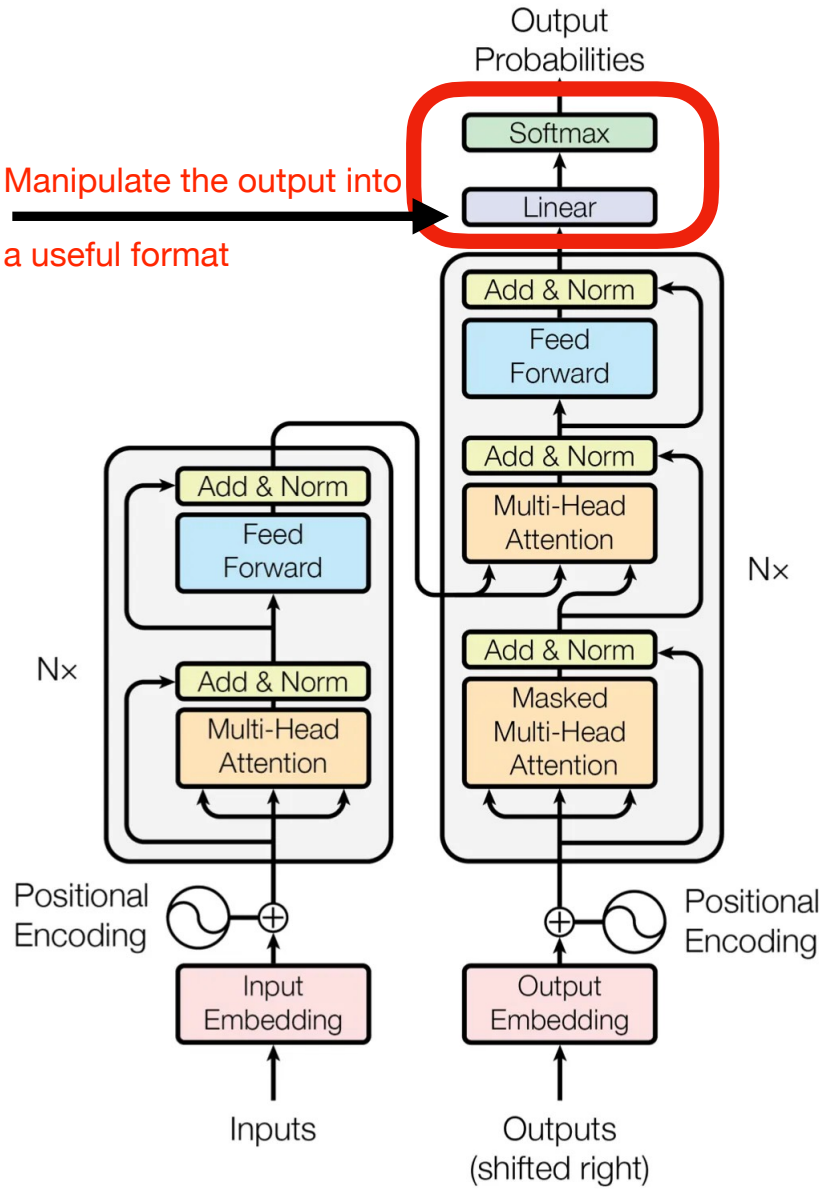
2. Helps model determine the relevant input sequence when generating the next token



1. The linear layer is another feedforward neural network, used again to manipulate the output into a useful format.

2. Softmax layer transforms the output into a probability distribution. This probability distribution is then used to predict the next word in the output sequence.

The whole process of this decoder block is repeated for each token in the output sequence, with the previously generated tokens being included as an input for each subsequent step.



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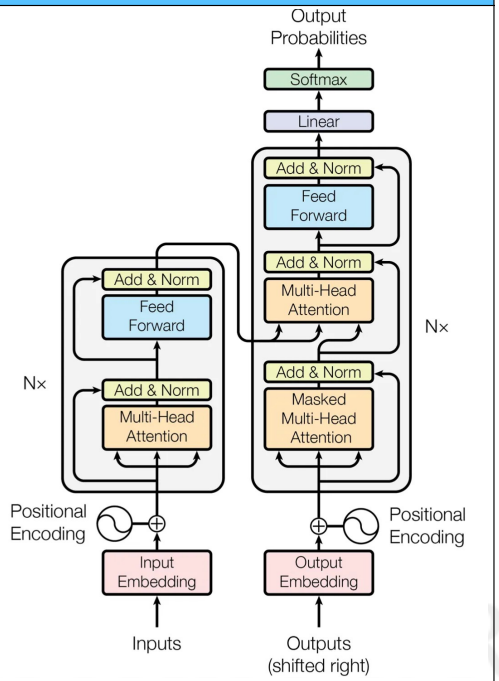
Chat GPT

Led by Dr. Shreyas Sarkar

Generative Pre-trained Transformer

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Generative	Pre-trained	Transformer
Can generate Human like Text	Learns from massive amount of data	
Continue the text in a coherent and contextually appropriate manner	Acquires deep understanding of language and grammar, and also of vocabulary	
Very good at : text completion, Dialogue Generation and Content Creation		

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